PROJECT PROBLEM STATEMENT

Chennai Metro Passenger Ridership and Urban Mobility Analysis Using Excel, MySQL and Power BI

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Course:	**Data Preprocessing and AI**
Project Title:	Chennai Metro Passenger Ridership and Urban Mobility Analysis Using Excel, MySQL and Power BI

Problem Statement

Chennai Metro passenger demand varies across stations, travel times, and ticketing methods. Analyzing these patterns can help identify high-demand stations, peak travel periods, and passenger ticketing preferences.

This project aims to analyze Chennai Metro ridership data using Excel, MySQL, and Power BI. The analysis will focus on station-wise passenger demand, hourly and monthly ridership trends, and the usage of different ticketing methods.

Objectives

1. Identify stations with the highest and lowest passenger demand.
2. Analyze hourly passenger traffic and peak travel periods.
3. Examine monthly ridership trends.
4. Understand passenger preferences for different ticketing methods.
5. Develop interactive Power BI dashboards to present insights.

Dataset

Chennai Metro (CMRL) Ridership & Network (2026) – Kaggle

Dataset files:

- cmrl_stationflow_daily.csv
- cmrl_hourly_ridership.csv
- cmrl_system_monthly.csv
- cmrl_ticket_mix.csv
- chennai_metro_stations.csv

Tools and Technologies

Microsoft Excel | **MySQL** | **Microsoft Power BI**

Expected Outcome

An interactive dashboard and data-driven insights highlighting Chennai Metro passenger demand, travel patterns, and ticketing preferences to support better service planning.